

VU Medical Zone

BT 405 Final Terms M.C.Q

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1. Which is the most important element for all living organisms in Earth? **Carbon**
2. Carbon-based nanomaterials demonstrate unprecedented physical and chemical properties such as: **high strength, resistance to corrosion and thermal conduction**
3. In graphite the carbon atom uses how many carbon atoms? **Three**
4. The allotrope of carbon atom named as buckyball was discovered in **1985**
5. The number of carbon atoms in buckyball is **60**
6. The diameter of Carbon nanotubes is ranges **1-3 nm**
7. Which one of the following are the properties of Carbon nanotubes? **Tensile Strength, Thermal Conductivity**
8. Some nanotubes have a property called _____, an expression of longitudinal twisting. **Chirality**
9. Multiple nanotubes can be assembled into microscopic mechanical systems called **Nanomachines**
10. If the nanotube structure is armchair then the electrical properties are **metallic**
11. The temperature stability of carbon nanotubes is estimated to be up to **2800°C** in vacuum and about **750°C** in air.
12. CNTs have not only unique atomic arrangements but also have unique properties that include large current carrying capability **high thermal conductivity, mechanical strength**
13. ESD is the full name of **Electrostatic discharge**
14. CNTs in _____ increases its tensile strength and halt crack propagation. **Concrete**
15. In Bucky paper thin nanotube sheets are _____ times stronger and _____ times lighter than steel. **250, 10**
16. Nanocomposites typically consist of _____ components. **Inorganic and Organic**
17. The field of ceramics that focuses on improving their mechanical properties is referred to as **structural ceramics**.
18. One of the most common reasons for adding fillers to polymers is to improve their **mechanical** performance.
19. _____ are a type of nanocomposite. **Nanocoatings**
20. A nanocoating refers to very fine, thin layers of polymeric chemical substances used to impart specific corrosion resistance, **chemical and physical** properties to a substrate surface.

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21. ALD stands for _____ **Atomic layer deposition**
22. _____ is the science and technology of interacting surfaces in relative motion. **Tribology**
23. _____ is often used as an alternative to nanocoatings. **Thin films**
24. Visible light is the part of the electromagnetic spectrum with wavelengths between _____ **400 and 700 nm.**
25. The resolving power of an optical microscope is around about _____ **0.2 μ m or 200 nm**
26. How many types of electron microscope? **Two**
27. STM is a Scanning Probe Microscopy (SPM) technique provides images of surfaces by scanning the surface line by line with a _____ **probe**
28. AFM is full form of _____ **Atomic force Microscopy**
29. Atomic force Microscopy measures the _____ force between the probe and the surface. **Attractive or repulsive**
30. _____ is defined as the branch of science that is concerned with the investigation and measurement of spectra produced when matter interacts with or emits electromagnetic (EM) radiation. **Spectroscopy**
31. X-rays were discovered in 1895 by German physicist _____ **Roentgen.**
32. A _____ is a device capable of recognizing a specific chemical species and signaling. **Sensor**
33. The _____ is a biomolecule that recognizes the target analyte. **Bioreceptor**
34. Proteins including enzymes and antibodies. Antibody based biosensors are also called _____ **Immunosensors**
35. _____ is capable of recognizing a specific target molecule. **Enzyme**
36. _____ should be capable of converting the biorecognition event into a measurable. **Transducer**
37. How many types of transducers are used to measure the Glucose concentration? **Three**
38. Nanoscale biosensors have the _____ to greatly aid in the diagnosis of diseases and monitoring of therapies. **Potential**
39. Which one are the basic characteristics of biosensor? **LINEARITY, SENSITIVITY and SELECTIVITY**
40. _____ is perhaps the most important feature of a biosensor. **Selectivity**
41. The best example of _____ is depicted by the interaction of an antigen with the antibody. **Selectivity**
42. _____ act as bioreceptors and are immobilised on the surface of the transducer. **Antibodies**

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43. _____ is the ability of the biosensor to generate identical responses for a duplicated experimental set-up.

Reproducibility

44. Linearity is the attribute that shows the _____ of the measured response.

Accuracy

45. The _____ of the biosensor is defined as the smallest change in the concentration of an analyte that is required to bring a change in the response of the biosensor.

Resolution

46. A _____ sensor converts light rays into an electronic signal.

Optical

47. **Photoconductive** devices used to measure the resistance by converting a change of incident light into a change of resistance.

48. The **photovoltaic cell (solar cell)** converts an amount of incident light into an output voltage.

49. The **Photodiodes** convert an amount of incident light into an output current.

50. _____ which involves measuring the current generated by electrochemical oxidation or reduction. **Amperometric**

51. _____ Used to detect the binding of biological molecules onto arrays of probe biomolecules covalently attached to chemically-modified gold surfaces. **Plasmonic biosensors**

52. The optical properties of noble metal nanoparticles are dominated by an effect called _____ **localized surface Plasmon resonance (SPR).**

53. One of the consequences of the LSPR effect in metal nanoparticles is that they have very _____ due to the resonant coherent oscillation of the plasmons. **Strong visible absorption**

54. In Amperometric Biosensor the substrate concentration is directly proportional to the _____ **magnitude of the current**

55. Which one of the following has example glucose by glucose (Blood glucose)? **Amperometric Biosensor**

56. _____ is the spread of cancer from one part of the body to another via either the circulatory system or lymphatic system. **Metastasis**

57. A molecule that specifically recognizes the analyte is known as _____ **bioreceptor**

58. In a biosensor the role of the transducer is to convert the bio-recognition event into a measurable signal. This process of energy conversion is known as _____ **signalization**

59. An immunosensor utilizes the very specific **binding affinity** of antibodies for a specific compound or antigen.

60. The use of _____ as the bio-recognition component of biosensors has several drawbacks. **Antibodies**

61. _____ are often used in bioreceptors because they are sensitive to surrounding environment. **Cell**

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62. Biosensors are devices comprising a biological element and a **physiochemical** detector that are used to detect analytes.
63. E. coli has been measured by detecting variation in pH caused by _____ **Ammonia**
64. Blood-glucose biosensors usage at home accounts for _____ of the gigantic world market. **85%**
65. Blood Glucose Monitoring is a way of checking the concentration of glucose in the blood using a **glucometer**
66. First-generation of glucose biosensors was first proposed by _____ **Clark and Lyons**
67. _____ one of the significant parameters used in the estimation of the concentration of biodegradable organic pollutants present in a water sample. **Biochemical oxygen demand**
68. _____ are widely used by food manufacturing industries as preservatives. **Nitrates**
69. _____ compounds are mainly used in the production of dyes, drugs, plastics, pesticides and detergents. **Phenolic**
70. _____ are a group of organic chemicals that are commonly used as insecticides, herbicides and pesticides in modern agriculture. **Organophosphorous (OP)**
71. Dioxins are polychlorinated compounds discharged as byproducts during several chemical processes involving **chlorine**.
72. Iron nanoparticles are _____ times more reactive than commonly used iron powders. **10 to 1 000**
73. MRI is the full form of _____ **Magnetic resonance imaging**
74. In the body, cells are supported in their growth and function by a natural scaffold called _____ **extracellular matrix (ECM)**
75. Neoplastic diseases include _____ **tumors and cancer**

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BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 64 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 1 of 10 (Start time: 01:07:43 AM, 19 August 2020)

Total Marks: 1

If the nanotube structure is armchair then the electrical properties will be -----?

Select the correct option

☒	Metallic	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Non-metallic	
☐	Neither metallic nor non- metallic	
☐	Convergent	

Click to Save Answer & Move to Next Question



BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 63 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 2 of 10 (Start time: 01:08:31 AM, 19 August 2020)

Total Marks: 1

Developed by the oil industry, MCM-41 with pore sizes in the range of _____ nanometers, is used for the removal of ultrafine contaminants

Select the correct option

☐	1 - 200	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	1 - 100	
☐	10 - 200	
☒	10 - 100	

Click to Save Answer & Move to Next Question



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BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 61 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 3 of 10 (Start time: 01:09:13 AM, 19 August 2020)

Total Marks: 1

Which of the following protein transports and stores iron in an organism, forms nanocavities----

Select the correct option

☐	Albumin	
☒	Ferritin	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Globulin	
☐	Ferricin	

Click to Save Answer & Move to Next Question



BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 54 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 4 of 10 (Start time: 01:09:58 AM, 19 August 2020)

Total Marks: 1

Block copolymers comprised of _____

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Select the correct option

☒	Hydrophobic & Hydrophilic block	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Organic blocks	
☐	Inorganic blocks	
☐	Crystalline block	

Click to Save Answer & Move to Next Question



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BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 22 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 5 of 10 (Start time: 01:10:54 AM, 19 August 2020)

Total Marks: 1

Which of the following physical parameter cannot be converted into an electrical signal by a sensor?

Select the correct option

☐	Temperature	
☐	Humidity	
☐	Speed	
☒	Antibodies	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 82 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 6 of 10 (Start time: 01:12:34 AM, 19 August 2020)

Total Marks: 1

The transduction methods used in biochips are _____.

Select the correct option

☐	Surface plasmon resonance	
☒	MRI	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	CT scan	
☐	Self Enhanced Resonance Spectroscopy	

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BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 65 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 7 of 10 (Start time: 01:12:51 AM, 19 August 2020)

Total Marks: 1

Polymers that are good conductors of electricity are called _____

Select the correct option

- | | | |
|----------------------------------|---------------------|---|
| ☐ | Platinum polymers | |
| ☒ | Conductive polymers | Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com |
| ☐ | Boron polymers | |
| ☐ | Block polymers | |

Click to Save Answer & Move to Next Question



BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 69 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 8 of 10 (Start time: 01:13:30 AM, 19 August 2020)

Total Marks: 1

More than _____ of biotech companies in the U.S. are involved in pharmaceutical production

Select the correct option

- | | | |
|----------------------------------|-----|---|
| ☐ | 60% | |
| ☐ | 50% | |
| ☐ | 70% | |
| ☒ | 65% | Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com |

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BC170406413: MUHAMMAD ALEEM ULLAH LIAQAT

Time Left 67 sec(s)

BT405:Quiz 01

Quiz Start Time: 01:07 AM, 19 August 2020

Question # 9 of 10 (Start time: 01:14:06 AM, 19 August 2020)

Total Marks: 1

The branch of science that deals with the chemical, physical and biological properties of surfaces is called

Select the correct option

☐	Physiology	
☐	Biochemistry	
☒	surface science	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Chemistry	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 1 of 10 (Start time: 02:16:16 AM, 19 August 2020)

Total Marks: 1

Which one of the following is not naturally occurring Zeolite?

Select the correct option

☒	Sodium	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Aluminum	
☐	Silicon	
☐	Calcium	

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 2 of 10 (Start time: 02:18:10 AM, 19 August 2020)

Total Marks: 1

A semiconductor whose excitons are confined in all three spatial dimensions is called as _____.

Select the correct option

☒	Qubit	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//
☐	Bucky ball		//
☐	Quantum dot		//
☐	Dot		//



Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 3 of 10 (Start time: 02:19:50 AM, 19 August 2020)

Total Marks: 1

Which of the following is application of Biosensors?

Select the correct option

☐	Clinical analysis and diagnosis of disease	//	
☐	Healthcare monitoring	//	
☐	Environmental pollution control	//	
☒	Measurement	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 4 of 10 (Start time: 02:20:43 AM, 19 August 2020)

Total Marks: 1

Nanotechnology can also be named as

Select the correct option

☐	Carbon Technology	
☒	Tissue Engineering	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Atomic Engineering	
☐	Atomic Physics	



Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 5 of 10 (Start time: 02:22:23 AM, 19 August 2020)

Total Marks: 1

Nanostructured materials differ from conventional polycrystalline materials in the ----- of the structural units?

Select the correct option

☐	Shape	
☒	Size	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Orientation	
☐	Quality	



Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 6 of 10 (Start time: 02:23:35 AM, 19 August 2020)

Total Marks: 1

Which one of the following is most rising application of the CNT?

Select the correct option

☒	Chemotherapy	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Solar Cells	
☐	Car batteries	
☐	Paper batteries	

[Click to Save Answer & Move to Next Question](#)



BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 7 of 10 (Start time: 02:25:48 AM, 19 August 2020)

Total Marks: 1

A nanometer is a unit of length in the metric system, equal to _____ of a meter

Select the correct option

☐	two millionth	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	one millionth	
☒	one billionth	
☐	two billionth	

[Click to Save Answer & Move to Next Question](#)



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BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 8 of 10 (Start time: 02:26:16 AM, 19 August 2020)

Total Marks: 1

DOXIL is;

Select the correct option

☐	Anti-bacterial drug	
☐	Anti-viral drug	
☐	Vaccine of Hepatitis	
☒	Anti-cancer drug	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 02:16 AM, 19 August 2020

Question # 9 of 10 (Start time: 02:26:30 AM, 19 August 2020)

Total Marks: 1

Carbon nanotubes are _____ times stronger than steel but _____ times lighter.

Select the correct option

☐	1000, ten	
☒	100, six	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	50, four	
☐	10, two	

Click to Save Answer & Move to Next Question



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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 21 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 1 of 10 (Start time: 02:00:52 AM, 19 August 2020)

Total Marks: 1

What does DNA microarray detect?

Select the correct option

☐	Position of genes	
☒	Expression of gene	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Number of genes	
☐	Sequence of genes	

Click to Save Answer & Move to Next Question



BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 57 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 2 of 10 (Start time: 02:02:34 AM, 19 August 2020)

Total Marks: 1

PET stands for _____

Select the correct option

☐	Polyethylene terephthalate	
☐	Polyethylene tetra sulphate	
☐	Plasmon Emission Tomography	
☒	Positron Emission Tomography	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 12 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 3 of 10 (Start time: 02:03:19 AM, 19 August 2020)

Total Marks: 1

The sheet of nanotubes is made up of _____

Select the correct option

☐	Aluminum	
☒	Graphene	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Copper	
☐	Silver	



Click to Save Answer & Move to Next Question



BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 67 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 4 of 10 (Start time: 02:05:07 AM, 19 August 2020)

Total Marks: 1

Diatoms are a type of _____ with nanostructured amorphous silica surfaces.

Select the correct option

☐	Fungi	
☒	Algae	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Bacteria	
☐	Polymers	

Click to Save Answer & Move to Next Question



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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 72 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 5 of 10 (Start time: 02:05:43 AM, 19 August 2020)

Total Marks: 1

STM

Select the correct option

☐	Resonance Tunneling Microscope	
☐	Resonance Tangent Microscope	
☐	Scanning Tangent Microscope	
☒	Scanning Tunneling Microscope	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 30 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 6 of 10 (Start time: 02:06:11 AM, 19 August 2020)

Total Marks: 1

What is grey goo-----?

Select the correct option

☒	A hypothetical substance composed of out-of-control self-replicating nanobots that consumes all living matter on Earth	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	The feeder material used to grow grey nanoparticles in the laboratory	
☐	Toxic byproduct resulting from the synthesis of carbon nanotubes	
☐	Waste product from the production of nanoglue made from the membranes on the feet of the Madagascar Grey Gecko	

Click to Save Answer & Move to Next Question



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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 66 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 7 of 10 (Start time: 02:07:34 AM, 19 August 2020)

Total Marks: 1

Quantum confinement causes the energy of the band gap to -----?

Select the correct option

☒	Increase	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//
☐	Decrease		//
☐	Remain Constant		//
☐	Variable		//

Click to Save Answer & Move to Next Question



BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 68 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 8 of 10 (Start time: 02:08:13 AM, 19 August 2020)

Total Marks: 1

----- technology is used for point-of-care testing in real time, which enhance delivery of medical care

Select the correct option

☐	Nanowire	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//
☐	Nanorobots		//
☒	Lab-on-chip		//
☐	Micro-lab		//

Click to Save Answer & Move to Next Question



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BC170406423: MUHAMMAD YASIR SHAHZAD

Time Left 17 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:00 AM, 19 August 2020

Question # 9 of 10 (Start time: 02:08:50 AM, 19 August 2020)

Total Marks: 1

An amount of incident light is converted into an output current by:

Select the correct option

☒	solar cell	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Photodiodes	
☐	Optical Sensors	
☐	Reflective Sensors	



Click to Save Answer & Move to Next Question

Windows taskbar showing icons for Internet Explorer, VLC, File Explorer, Firefox, Adobe Reader, and Chrome. System tray shows 2:10 AM, 19-Aug-20, and network status.

BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 1 of 10 (Start time: 01:32:54 AM, 19 August 2020)

Total Marks: 1

One of the significant parameter used in the estimation of the concentration of biodegradable organic pollutants present in a water sample is-----?

Select the correct option

☐	Biological oxygen demand	
☒	Biochemical oxygen demand	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Chemical oxygen demand	
☐	Catalytic oxidation	

Click to Save Answer & Move to Next Question

Windows taskbar showing icons for Internet Explorer, VLC, File Explorer, Firefox, Adobe Reader, and Chrome. System tray shows 1:33 AM, 19-Aug-20, and network status.

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BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 2 of 10 (Start time: 01:33:54 AM, 19 August 2020)

Total Marks: 1

The branch of science that is concerned with the investigation and measurement of spectra produced when matter interacts with or emits electromagnetic (EM) radiation

Select the correct option

☐	Radiology	
☒	Spectroscopy	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Electroradioscopy	
☐	None of the given options	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 3 of 10 (Start time: 01:34:29 AM, 19 August 2020)

Total Marks: 1

Quantum dots are _____ in nature.

Select the correct option

☒	Inorganic	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Organic	
☐	Biologic	
☐	Metallic	

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 4 of 10 (Start time: 01:36:26 AM, 19 August 2020)

Total Marks: 1

DOXIL is;

Select the correct option

- | | | |
|----------------------------------|----------------------|--|
| ☐ | Anti-bacterial drug | |
| ☐ | Anti-viral drug | |
| ☐ | Vaccine of Hepatitis | |
| ☒ | Anti-cancer drug | Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com |

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 5 of 10 (Start time: 01:37:05 AM, 19 August 2020)

Total Marks: 1

The substance used in combination with a specific antigen that produced a more robust immunity than the antigen alone is -----?

Select the correct option

- | | | |
|----------------------------------|------------|--|
| ☐ | Propagator | |
| ☒ | Adjuvant | Answer solved by Amaan Khan
Contact: 0305-4716616
Email: maniamaan2@gmail.com |
| ☐ | Microbead | |
| ☐ | Emulsifier | |

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 6 of 10 (Start time: 01:38:52 AM, 19 August 2020)

Total Marks: 1

The quantum confinement effect is observed when the size of the nanoparticle is ----- to be comparable to the wavelength of the electron.

Select the correct option

☐	large	
☒	small	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	constant	
☐	equal to wavelength of electron	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 7 of 10 (Start time: 01:39:41 AM, 19 August 2020)

Total Marks: 1

The temperature stability of nanotubes in vacuum is

Select the correct option

☐	1900 °C	
☐	2300 °C	
☒	2800 °C	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	3200 °C	

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 8 of 10 (Start time: 01:40:13 AM, 19 August 2020)

Total Marks: 1

Longitudinal twisting is an expression of a property of carbon nanotube which is -----?

Select the correct option

☒	Chirality	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//
☐	Blending		//
☐	Rotation		//
☐	Both B and C		//

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 01:32 AM, 19 August 2020

Question # 9 of 10 (Start time: 01:40:59 AM, 19 August 2020)

Total Marks: 1

The most common nanoscale substance used in consumer products is _____

Select the correct option

☒	Nano-silver	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com	//
☐	Nano-chlorine		//
☐	Nano-cupper		//
☐	Nano-Zinc		//

Click to Save Answer & Move to Next Question



EFFORT BY AMAAN KHAN

VU Medical Zone

BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 1 of 10 (Start time: 10:42:17 PM, 18 August 2020)

Total Marks: 1

Expand ROS

Select the correct option

☐	Responsive oxygen species	
☒	Reactive oxygen species	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Receptive oxygen species	
☐	Responsible oxygen species	



Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 2 of 10 (Start time: 10:43:36 PM, 18 August 2020)

Total Marks: 1

Which one of the following is neoplastic disease?

Select the correct option

☐	Hypertension	
☐	Coronary disease	
☒	Cancer	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Parkinson's disease	

Click to Save Answer & Move to Next Question



EFFORT BY AMAAN KHAN

VU Medical Zone

BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 3 of 10 (Start time: 10:44:17 PM, 18 August 2020)

Total Marks: 1

The transduction methods used in biochips are _____.

Select the correct option

☐	Surface plasmon resonance	
☐	MRI	
☒	CT scan	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Self Enhanced Resonance Spectroscopy	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 4 of 10 (Start time: 10:46:13 PM, 18 August 2020)

Total Marks: 1

Which of the following is natural self adhesive material?

Select the correct option

☐	Silver nanodots	
☐	Nanobeads	
☐	Nanocavities	
☒	Diatoms	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



VU Medical Zone

BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 5 of 10 (Start time: 10:47:47 PM, 18 August 2020)

Total Marks: 1

A sensor which converts light rays into an electronic signal is:

Select the correct option

☒	Optical sensors	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Sensors	
☐	Biosensors	
☐	Bioreceptors	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 6 of 10 (Start time: 10:48:18 PM, 18 August 2020)

Total Marks: 1

Which of the following isn't a benefit of nanocoatings?

Select the correct option

☐	Substrate waterproofing	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Prolonged substrate lifespan and application use	
☐	Ultraviolet (UV) stability	
☒	Urea preparation	

Click to Save Answer & Move to Next Question



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BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 7 of 10 (Start time: 10:49:17 PM, 18 August 2020)

Total Marks: 1

Who used the word Nanotechnology for the first time?

Select the correct option

☐	Professor Claudio of SEPRL, USA	
☒	Professor Taniguchi of Tokyo Science University	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Professor Patti Miller of Japan Science University	
☐	Professor David of Japan Science University	



Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 8 of 10 (Start time: 10:50:25 PM, 18 August 2020)

Total Marks: 1

A molecule that specifically recognizes the analyte is known as -----?

Select the correct option

☐	Biomarker	
☒	Bioreceptor	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Biomachine	
☐	Transducer	

Click to Save Answer & Move to Next Question



EFFORT BY AMAAN KHAN

VU Medical Zone

BT405:Quiz 01

Quiz Start Time: 10:42 PM, 18 August 2020

Question # 9 of 10 (Start time: 10:51:31 PM, 18 August 2020)

Total Marks: 1

Smart coatings can also be passive in the sense of changing their optical properties due to:

Select the correct option

☒	Thermochromic & photochromic	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	wavelength	
☐	infrared	
☐	environmental conditions	

Click to Save Answer & Move to Next Question



BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 1 of 10 (Start time: 02:23:21 AM, 18 August 2020)

Total Marks: 1

The quantum confinement effect is observed when the size of the nanoparticle is ----- to be comparable to the wavelength of the electron.

Select the correct option

☐	large	
☒	small	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	constant	
☐	equal to wavelength of electron	

Click to Save Answer & Move to Next Question



EFFORT BY AMAAN KHAN

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BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 2 of 10 (Start time: 02:24:26 AM, 18 August 2020)

Total Marks: 1

Which of the following is not an example of nanostructured materials?

Select the correct option

☐	Nanocrystalline	
☐	Nanocomposite	
☐	Hybrid materials	
☒	All of these are nanostructured materials	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



BC170404600: SHAHBAZ ALI

Time Left 12 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 3 of 10 (Start time: 02:25:54 AM, 18 August 2020)

Total Marks: 1

Different phases of liquid crystals can be distinguished by their

Select the correct option

☐	Electric properties	
☐	Magnetic properties	
☐	Chemical properties	
☒	Optical properties	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com

Click to Save Answer & Move to Next Question



EFFORT BY AMAAN KHAN

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BC170404600: SHAHBAZ ALI

Time Left 42 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 4 of 10 (Start time: 02:27:44 AM, 18 August 2020)

Total Marks: 1

Diameter of carbon nanotubes is

Select the correct option

☐	<100 nm	
☐	<50 nm	
☒	30 nm	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	None of the given options	



Click to Save Answer & Move to Next Question



BC170404600: SHAHBAZ ALI

Time Left 54 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 5 of 10 (Start time: 02:28:58 AM, 18 August 2020)

Total Marks: 1

The material which contains two or more phase separated components with morphology of spheres, cylinders or networks with nano-sized dimensions is -----?

Select the correct option

☐	Nanocylinder	
☐	Nanostructure	
☒	Nanocomposite	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Nanodot	

Click to Save Answer & Move to Next Question



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BC170404600: SHAHBAZ ALI

Time Left 11 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 6 of 10 (Start time: 02:29:50 AM, 18 August 2020)

Total Marks: 1

Which process is mainly used in the formation of Nanofibers?

Select the correct option

☐	Electroplating	
☐	Spectrometry	
☐	Cytometry	
☒	Electrospinning	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com



Click to Save Answer & Move to Next Question



BC170404600: SHAHBAZ ALI

Time Left 45 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 7 of 10 (Start time: 02:31:43 AM, 18 August 2020)

Total Marks: 1

The self-healing properties of natural adhesives are due to the properties of the ----- contained in them?

Select the correct option

☒	Proteins	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Glycans	
☐	minerals	
☐	covalent bond	

Click to Save Answer & Move to Next Question



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BC170404600: SHAHBAZ ALI

Time Left 46 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 8 of 10 (Start time: 02:32:48 AM, 18 August 2020)

Total Marks: 1

Antibody-based biosensors are also known as_____.

Select the correct option

☐	Microbiosensors	
☒	Immunosensors	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Potentiometric Biosensors	
☐	Biochips	

Click to Save Answer & Move to Next Question



BC170404600: SHAHBAZ ALI

Time Left 7 sec(s)

BT405:Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 9 of 10 (Start time: 02:33:55 AM, 18 August 2020)

Total Marks: 1

----- is used to avoid nanoparticles from agglomeration and they don't get deposited.

Select the correct option

☒	Surfactant	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Algal Extract	
☐	Ceramic	
☐	Choloroform	

Click to Save Answer & Move to Next Question



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BC170404600: SHAHBAZ ALI

Time Left 60 sec(s)

BT405: Quiz 01

Quiz Start Time: 02:23 AM, 18 August 2020

Question # 10 of 10 (Start time: 02:35:47 AM, 18 August 2020)

Total Marks: 1

The drug resistance occurs due to the presence of

Select the correct option

☐	Chemical barrier	
☒	Physiological barrier	Answer solved by Amaan Khan Contact: 0305-4716616 Email: maniamaan2@gmail.com
☐	Physiochemical barrier	
☐	None of these	

Click to Save Answer & Move to Next Question



SUBJECTIVE

Bt405

What is biochip?

2 marks

- There are different names for the microarrays, like DNA/RNA Chips, BioChips or GeneChips.

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- The array can be defined as an ordered collection of microspots, each spot containing a single defined species of a nucleic acid.
- The microarray technique is based on hybridisation of nucleic acids.
- There exist two variants of the chips:

cDNA microarrays

oligonucleotide arrays

Properties of ceramics and glass

3 marks

- Ceramic materials by definition are ionically bound; they are hard materials, both electrically and thermally very stable.
- Included in this category are, for example, Al_2O_3 , Si_3N_4 , MgO , SiO_2 , Na_2O , CaO and ZrO_2 .
- Ceramics are characterised by being hard yet brittle, therefore, in many cases, they are used as composites where they are mixed with other materials (e.g. metals) to increase their mechanical performance.

Properties:

- High optical transparencies.
- Silica aerogels are made of pores of about 10 nm arranged in distances between 10 and 100 nm.
- They are resistant and chemically inert to liquid metals, heat resistant up to 1 200 °C and non-toxic.
- Biomedical applications, such as substrates for cell growth and analysis.

Features of sensors

3 marks

- A sensor is a device capable of recognizing a specific chemical species and signaling the presence, activity or concentration of that species in solution through some chemical change.

Feature of sensor:

- Biosensor = bioreceptor + transducer.
- A device that utilizes biological components e.g. enzymes to indicate the amount of a biomaterial.
- A biosensor is an analytical device, used for the detection of an analyte, that combines a biological component with a physicochemical detector.
- The bioreceptor is a biomolecule that recognizes the target analyte whereas the transducer converts the recognition event into a measurable signal.

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➤ The sensitive biological element

(e.g. tissue, microorganisms, organelles, cell receptors, enzymes, antibodies, nucleic acids, etc.)

is a biologically derived material or biomimetic component that interacts (binds or recognizes) with the analyte under study.

Application of biosensor in environment

3 marks

With high specificity and sensitivity, biosensors provide an exceptional analytical system for the monitoring of environmental pollutants.

They are inexpensive and attractive alternative to the conventional analytical techniques, which are capable of providing real-time online monitoring.

With the diverse effect of these pollutants on the biological system, a number of biosensors have been developed and are still in progress.

Among toxic compounds, determination of heavy metals, phenolic compounds, mercury, organophosphorus, and carbamate pesticides is the major concern, considering their extensive contribution in increased pollutant level.

Biosensor or biodevices can be used as environmental quality monitoring tools in the assessment of physical, chemical and biological monitoring of pollutants.

The major applications of biosensors are monitoring of various pollutants including heavy metals, organic and inorganic pollutants, toxins, antibiotics and contaminating microorganisms.

Heavy metals

Heavy metals, i.e. copper (Cu), cadmium (Cd), mercury (Hg), lead (Pb), zinc (Zn), etc., are recently the major cause of serious environmental pollution problems.

These are known for their high toxicity and bioaccumulation attribute in the food chain.

A number of bacterial biosensors have been developed for the determination of heavy metals in different environmental samples using enzyme and DNA as bioreceptor and optical and electrochemical as transduction system.

Biochemical oxygen demand (BOD)

Biochemical oxygen demand is one of the significant parameters used in the estimation of the concentration of biodegradable organic pollutants present in a water sample

Nitrogen compounds

Nitrogen compounds, i.e. nitrates, are widely used by food manufacturing industries as preservatives (increase shelf-life) and chemical fertilizer industries as fertilizer (increase the fertility of the soil).

Polychlorinated biphenyls (PCBs)

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PCBs are toxic organic compounds, universal environment pollutants; hence several countries banned their production long time ago.

Such compounds are highly lypophilic nature, therefore abundant chances of accumulating in the food chain.

Phenolic compounds

Phenols and their derivatives, i.e. chlorophenol are distributed commonly in the environment.

Such compounds are mainly used in the production of dyes, drugs, plastics, pesticides, detergents, etc..



Detail on bioreceptors.

5 marks

Biosensors:

The sensor which shows the reaction in efficiency is called biosensor. Biosensor can be enzyme. In other words enzymes can act as a biosensor. It is a device which makes the detection of analyte which connects to a biological part By a detector whose is physiochemical. The sensitive elements can be made by biological engineering. This kind of detector elements are very useful and known as biosensors. Biosensor reader are the devices which are electronically joint with biosensor.

Biosensor system:

Biosensors are consisted of two site one is recognition site which is also called bioreceptors. Recognition component is use the biomolecules for the interaction with analyte which is interested for the wok. Main purpose of biosensor is to make efficient suitable care.

Biosensor types:

- Electronic biosensors
- Pyroelectric biosensor
- Gravimetric biosensor
- Optical biosensors
- Electrochemical

Uses:

To quantify the glucose focus, three unique transducers can be utilized:

- An oxygen sensor that measures oxygen fixation
- A pH sensor that measures the corrosive (gluconic corrosive) creation
- A peroxide sensor that measures H₂O₂ fixation.
- An oxygen sensor is a transducer that believers oxygen fixation into electrical current.
- pH sensor is a transducer that believers pH change into voltage change.

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- A peroxidase sensor is a transducer that proselytes peroxidase fixation into an electrical current.

Cholesterol monitoring:

1. Low motivating force to utilize
2. Cost high
3. What do you do with the data?
4. Learnt nothing from the Glucose ventures improvement.

Contemplations in Biosensor

Once an objective analyte has been recognized, the real assignments in building up a biosensor includes:

- Selection of an appropriate bioreceptor atom
- Selection of an appropriate immobilization strategy
- Selection of an appropriate transducer
- Designing of biosensor considering estimation range, linearity, and minimization of obstruction
- Packaging of biosensor

Applications of biosensors:

Biosensors are gadgets including a natural component and a physiochemical indicator that are utilized to distinguish analytes. These instruments have an extensive variety of utilizations running from clinical through to natural and farming. The gadgets are likewise utilized as a part of the sustenance business. A few cases of the fields that utilization biosensor innovation include:

- General medicinal services observing
- Screening for ailment
- Clinical investigation and analysis of sickness
- Veterinary and agrarian applications
- Modern preparing and checking
- Ecological contamination control
- In sustenance preparing, checking, nourishment genuineness, quality and security
- Quality and security, support of nourishment items and handling

Nourishment verification and observing with objective and steady estimation of sustenance items, in a financially savvy way, are attractive for the nourishment business.

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Therefore improvement of biosensors in light of the interest for basic, ongoing, particular and reasonable procedures is apparently auspicious. Observing of maturing of lager utilizing enzymatic biosensors, in light of cobalt phthalocyanine – amid capacity .Biosensors are utilized for the location of pathogens in sustenance.

Nearness of Escherichia coli in vegetables, is a bioindicator of fecal sully in sustenance. E. coli has been estimated by identifying variety in pH caused by smelling salts (created by urease– E. coli immune response conjugate) utilizing potentiometric rotating biosensing frameworks.

In maturation forms :

In maturation businesses, process wellbeing and item quality are essential.

Along these lines viable checking of the maturation procedure is basic to create, upgrade and keep up natural reactors at greatest viability. Biosensors can be used to screen the nearness of items, biomass, protein, counter acting agent or side-effects of the procedure to in a roundabout way measure the procedure conditions. Biosensors unequivocally control the maturation business and deliver reproducible outcomes because of their basic instrumentation, impressive selectivity, low costs and simple robotization. Biosensing innovation for manageable sustenance wellbeing The term nourishment quality alludes to the appearance, taste, smell, wholesome esteem, freshness, flavor, surface and synthetic concoctions.

Savvy observing of supplements and quick screening of organic and synthetic contaminants are of principal significance, with regards to nourishment quality and security. Glutamine is the quick and dirty of essential capacities, for example. Patients insufficient in glutamine experience the ill effects of pathologies, for example, malabsorptive clutters and must be supplemented, to enhance insusceptible capacities. Glutaminase-based microfluidic biosensor chip with a stream infusion examination for electrochemical recognition has been utilized for location in maturation process.

In restorative field:

In the teach of restorative science, the utilizations of biosensors are developing quickly. Glucose biosensors are generally utilized as a part of clinical applications for analysis of diabetes mellitus, which requires exact control over blood-glucose levels. Blood-glucose biosensors use at home records for 85% of the colossal world market. Biosensors are being utilized unavoidably in the restorative field to analyze irresistible sicknesses. A promising biosensor innovation for urinary tract disease (UTI) conclusion alongside pathogen distinguishing proof and against microbial powerlessness is under examination. A novel biosensor, in light of hafnium oxide (HfO₂), has been utilized for beginning time discovery of human interleukin (IL).

With high specificity and sensitivity, biosensors provide an exceptional analytical system for the monitoring of environmental pollutants.

They are inexpensive and attractive alternative to the conventional analytical techniques, which are capable of providing real-time online monitoring.

With the diverse effect of these pollutants on the biological system, a number of biosensors have been developed and are still in progress.

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Among toxic compounds, determination of heavy metals, phenolic compounds, mercury, organophosphorus, and carbamate pesticides is the major concern, considering their extensive contribution in increased pollutant level.

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Phenolic compounds

Phenols and their derivatives, i.e. chlorophenol are distributed commonly in the environment.

Such compounds are mainly used in the production of dyes, drugs, plastics, pesticides, detergents, etc..

IN food industry:

- Food processing industry faces various challenges; one of the foremost challenges is the need for quick and cost effective methods to detect the presence of allergenic components and pathogens in the food.
- Biosensors pave way for the rapid detection of pathogens, allergens as well as pesticide residues in food.

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- Detection of contaminants, verification of product contents, product freshness and monitoring of raw materials conversion are the areas of potential biosensor applications.
- Biosensors have the potential to produce an analytical revolution to resolve the challenges in the agricultural and the food industries.
- The determination of chemical and biological contaminants in foods is of paramount importance to the health of food because, unlike the contamination of a physical nature, they cannot be displayed.
- Development of biosensors to the analysis of the quality of food, since they have proven to be an extremely viable alternative to traditional analytical techniques such as chromatography.
- In the area of food the interest in the development of biosensors mainly focuses on analysis of food security (detection of compound contaminants, allergens, toxins, pathogens, and additives etc.) Food composition and online process control.

Biosensors for diabetes applications

Glucose as diabetes biomarker

- About 3% of the population worldwide suffers from diabetes, a leading cause of death, and its incidence is growing fast.
- Diabetes is a syndrome of disordered metabolism resulting in abnormally high blood sugar levels.
- Optimal management of diabetes involves patients measuring and recording their own blood glucose levels.
- Under normal physiological condition, the concentration of fasting plasma glucose is in the range 6.1–6.9 mmolL⁻¹, so the variation of the blood glucose level can indicate diabetes mellitus, besides other conditions.
- Association recommends that insulin-dependent type 1 diabetics self monitor blood glucose 3–4 times daily, while insulin-dependent type 2 diabetics monitor once-daily.
- Therefore it is necessary to develop a simple, sensitive, accurate, micro-volume and
- low-cost approach for glucose analysis which is appropriate for rapid field tests and is also effective as an alternative to the existing methods.

First-generation of glucose biosensors (Clark enzyme electrode)

- First proposed in 1962 by Clark and Lyons
- These sensors were initially based on an electrochemical approach, which used the enzyme glucose oxidase (GOx).
- Electrochemical sensors were chosen for blood-glucose measurements due to their high sensitivity, on the order of μM to mM, good reproducibility and ease of fabrication at relatively low cost.

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- GOx catalyses the oxidation of glucose to gluconolactone in the presence of oxygen, while producing hydrogen peroxide (H_2O_2) and water as by-products.
- Measurements of peroxide formation have the advantage of being simpler, especially when miniature devices are being considered.
- The first commercially successful glucose biosensor using Clark's technology was the Yellow Springs Instrument Company analyzer (Model 23A YSI analyzer) for the direct measurement of glucose in 1975, which was based on the amperometric detection of hydrogen peroxide.
- This analyzer was almost exclusively used in clinical laboratories because of its high cost due to the expensive platinum electrode.

Explain self adhesive and why diatoms are good model for biomimetic membrane 5

Self-healing adhesives

- Diatoms are a type of algae with nanostructured amorphous silica surfaces.
- Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.
- Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO_2).
- Diatoms are often among the first organisms to colonize underwater surfaces, because their adhesive mucilage enables the cells to attach to substrata of a very wide range of chemical composition (organic, inorganic, natural or man-made).
- Surface adhesion of diatoms belonging to the raphe pennate type requires the secretion of so-called adhesive mucilage strands through a dedicated slit in the silica cell wall, which is termed the raphe.

Self-healing adhesives

- Adhering cells can then move across the surface, which is accomplished through an actinmyosin dependent cytoskeletal motor that translocates the adhesive mucilage strands in a rearward direction through the raphe, thereby moving the cell forward.
- Some diatom species have evolved strong self-healing underwater adhesives. Some are free-floating, others have adhesive properties in water: for instance, diatoms in the Antarctic seas can attach to ice.
- Others secrete viscous mucilage which binds colonies together while protecting the silica shells from wear as they rub against each other.
- Molluscs are another species that have adhesive properties under water.

- Both diatoms and molluscs have strong underwater glues that can also resist stress and self-heal.

They serve as a biomimetic model for self-healing materials.

Self-healing adhesives

- These natural adhesives and their self-healing properties are due to the properties of the proteins contained in them.
- These proteins have sacrificial bonds that allow the molecule to be reversibly stretched by breaking and re-bonding.
- This sacrificial bond behaviour has been observed in many other materials, such as wool. Closing the wounds with sticky materials has been introduced to prevent bleeding and induce the wound healing process.
- Many types of surgical adhesives including tissue adhesives have been developed to be a suitable alternative for sutures and staples.
- A new approach which aims at producing bio-adhesives by mimicking the nature along with applying nanotechnology methods.

Self-healing adhesives

- Engineering stable underwater adhesives currently poses a major technical challenge, because most man-made adhesives fail in wet conditions, owing to chemical modification of the adhesive or its substrate.
- Diatoms produce adhesives that are extremely strong and robust, in both fresh- and seawater environments.
- AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.
- The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoelastic properties, adhesion forces and hardness measurements, in physiological conditions.
- Experiments with three diatom species that produce outstanding natural adhesives: *Eunotia sudetica*, *Navicula seminulum* and an unidentified species

What is spectroscopy and write X-ray method in detail 5

Spectroscopy is defined as the branch of science that is concerned with the investigation and measurement of spectra produced when matter interacts with or emits electromagnetic (EM) radiation.

Spectroscopy is a technique that uses the interaction of energy with a sample to perform an analysis.

Depending on the wavelength of the electromagnetic used and the type of interaction with matter that occurs (absorption, scattering, etc.), different spectra are measured from which much information can be inferred

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X ray:

X-rays were discovered in 1895 by German physicist Roentgen.

X-ray methods involve exciting a sample either with X-rays (creating more X-rays) or with electrons (creating X-rays).

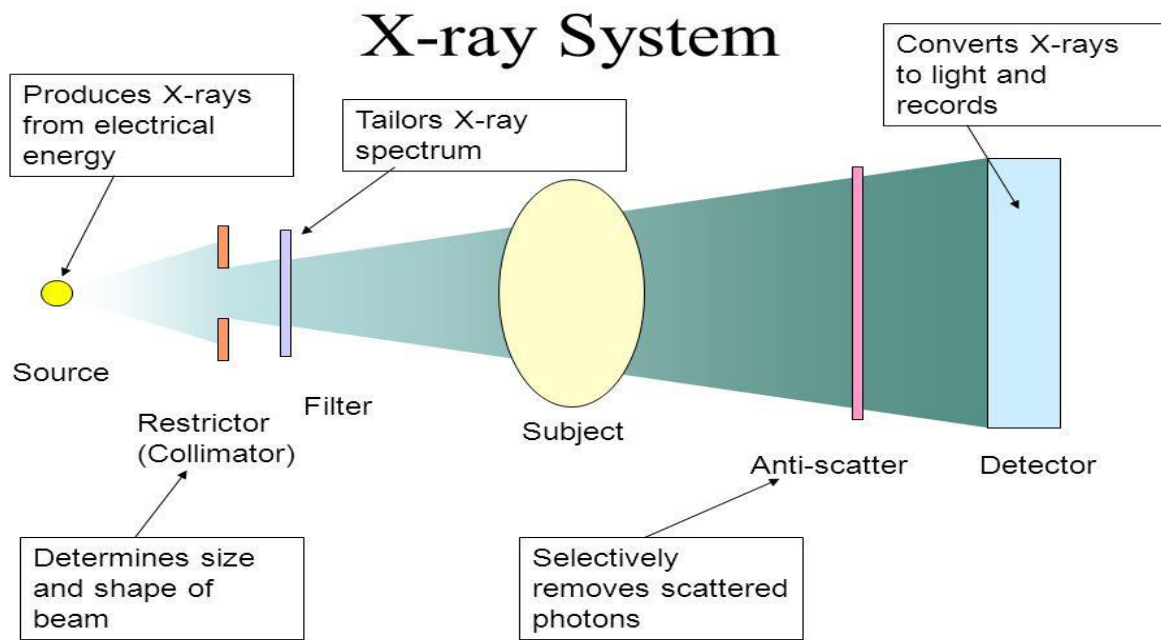
X-rays can be also generated by bombarding a sample with alpha particles.

The energy of emitted X-rays is equal to the difference between the binding energies of the electrons involved in the transition.

There are various methods that use X-rays: X-ray fluorescence (XRF), X-ray diffraction (XRD), etc.

In the context of nanomaterials, the most important method is small-angle X-ray scattering (SAXS) analysis.

Like XRD, this method is based on the principle of scattering of X-rays.



Spanning a wide range of research fields, such as biology, chemistry, physics, and engineering, nanotechnology is uniquely interdisciplinary in character.

The common factor in nanotechnology is the lateral dimension of the structures studied.

Defined as materials being in, or having components in, the one billionth (10^{-9}) of a meter range, nanotechnology research and development relies on accurate measurement of atomic and molecular distances within structures ranging from semiconductor devices to nano-powders.

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The dimensions of X-ray wavelengths are of the same magnitude as the size of nanostructures, X-ray diffraction (XRD) and associated techniques are primary tools for the nanotechnology researcher.

X-ray reflectometry (XRR) determines layer thickness, roughness, and density.

High-resolution X-ray diffraction can measure layer thickness, roughness, chemical composition, lattice spacing, relaxation and more.

X-ray diffuse scattering is used to determine lateral and transversal correlations, distortions, density, and porosity.

A new advance in X-ray imaging has revealed the dramatic three-dimensional shape of gold nanocrystals, and is likely to shine a light on the structure other nano-scale materials.

The new technique improves the quality of nanomaterial images, made using X-ray diffraction, by accurately correcting distortions in the X-ray light.

Most nanomaterial imaging has been done using electron microscopy. X-ray imaging is an attractive alternative as X-rays penetrate further into the material than electrons and can be used in ambient or controlled environments.

Detail on application of biosensors 10 marks

Biosensors:

The sensor which shows the reaction in efficiency is called biosensor. Biosensor can be enzyme. In other words enzymes can act as a biosensor. It is a device which makes the detection of analyte which connects to a biological part By a detector whose is physiochemical. The sensitive elements can be made by biological engineering. This kind of detector elements are very useful and known as biosensors. Biosensor reader are the devices which are electronically joint with biosensor.

Biosensor system:

Biosensors are consisted of two site one is recognition site which is also called bioreceptors. Recognition component is use the biomolecules for the interaction with analyte which is interested for the wok. Main purpose of biosensor is to make efficient suitable care.

Biosensor types:

- Electronic biosensors
- Pyroelectric biosensor
- Gravimetric biosensor
- Optical biosensors
- Electrochemical

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Uses:

To quantify the glucose focus, three unique transducers can be utilized:

- An oxygen sensor that measures oxygen fixation
- A pH sensor that measures the corrosive (gluconic corrosive) creation
- A peroxide sensor that measures H₂O₂ fixation.
- An oxygen sensor is a transducer that believers oxygen fixation into electrical current.
- pH sensor is a transducer that believers pH change into voltage change.
- A peroxidase sensor is a transducer that proselytes peroxidase fixation into an electrical current.

Cholesterol monitoring:

1. Low motivating force to utilize
2. Cost high
3. What do you do with the data?
4. Learnt nothing from the Glucose ventures improvement.

Contemplations in Biosensor Development

Once an objective analyte has been recognized, the real assignments in building up a biosensor includes:

- Selection of an appropriate bioreceptor atom
- Selection of an appropriate immobilization strategy
- Selection of an appropriate transducer
- Designing of biosensor considering estimation range, linearity, and minimization of obstruction
- Packaging of biosensor

Applications of biosensors:

Biosensors are gadgets including a natural component and a physiochemical indicator that are utilized to distinguish analytes. These instruments have an extensive variety of utilizations running from clinical through to natural and farming. The gadgets are likewise utilized as a part of the sustenance business. A few cases of the fields that utilization biosensor innovation include:

- General medicinal services observing
- Screening for ailment
- Clinical investigation and analysis of sickness
- Veterinary and agrarian applications

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- Modern preparing and checking
- Ecological contamination control
- In sustenance preparing, checking, nourishment genuineness, quality and security
- Quality and security, support of nourishment items and handling

Nourishment verification and observing with objective and steady estimation of sustenance items, in a financially savvy way, are attractive for the nourishment business.

Therefore improvement of biosensors in light of the interest for basic, ongoing, particular and reasonable procedures is apparently auspicious. Observing of maturing of lager utilizing enzymatic biosensors, in light of cobalt phthalocyanine – amid capacity .Biosensors are utilized for the location of pathogens in sustenance.

Nearness of Escherichia coli in vegetables, is a bioindicator of fecal sully in sustenance. E. coli has been estimated by identifying variety in pH caused by smelling salts (created by urease– E. coli immune response conjugate) utilizing potentiometric rotating biosensing frameworks.

In maturation forms :

In maturation businesses, process wellbeing and item quality are essential.

Along these lines viable checking of the maturation procedure is basic to create, upgrade and keep up natural reactors at greatest viability. Biosensors can be used to screen the nearness of items, biomass, protein, counter acting agent or side-effects of the procedure to in a roundabout way measure the procedure conditions. Biosensors unequivocally control the maturation business and deliver reproducible outcomes because of their basic instrumentation, impressive selectivity, low costs and simple robotization. Biosensing innovation for manageable sustenance wellbeing The term nourishment quality alludes to the appearance, taste, smell, wholesome esteem, freshness, flavor, surface and synthetic concoctions.

Savvy observing of supplements and quick screening of organic and synthetic contaminants are of principal significance, with regards to nourishment quality and security. Glutamine is the quick and dirty of essential capacities, for example. Patients insufficient in glutamine experience the ill effects of pathologies, for example, malabsorptive clutters and must be supplemented, to enhance insusceptible capacities. Glutaminase-based microfluidic biosensor chip with a stream infusion examination for electrochemical recognition has been utilized for location in maturation process.

With high specificity and sensitivity, biosensors provide an exceptional analytical system for the monitoring of environmental pollutants.

They are inexpensive and attractive alternative to the conventional analytical techniques, which are capable of providing real-time online monitoring.

With the diverse effect of these pollutants on the biological system, a number of biosensors have been developed and are still in progress.

Among toxic compounds, determination of heavy metals, phenolic compounds, mercury, organophosphorus, and carbamate pesticides is the major concern, considering their extensive contribution in increased pollutant level.

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Biosensor or biodevices can be used as environmental quality monitoring tools in the assessment of physical, chemical and biological monitoring of pollutants.

The major applications of biosensors are monitoring of various pollutants including heavy metals, organic and inorganic pollutants, toxins, antibiotics and contaminating microorganisms.

Heavy metals

Heavy metals, i.e. copper (Cu), cadmium (Cd), mercury (Hg), lead (Pb), zinc (Zn), etc., are recently the major cause of serious environmental pollution problems.

These are known for their high toxicity and bioaccumulation attribute in the food chain.

A number of bacterial biosensors have been developed for the determination of heavy metals in different environmental samples using enzyme and DNA as bioreceptor and optical and electrochemical as transduction system.

Biochemical oxygen demand (BOD)

Biochemical oxygen demand is one of the significant parameters used in the estimation of the concentration of biodegradable organic pollutants present in a water sample

Nitrogen compounds

Nitrogen compounds, i.e. nitrates, are widely used by food manufacturing industries as preservatives (increase shelf-life) and chemical fertilizer industries as fertilizer (increase the fertility of the soil).

Polychlorinated biphenyls (PCBs)

PCBs are toxic organic compounds, universal environment pollutants; hence several countries banned their production long time ago.

Such compounds are highly lipophilic nature, therefore abundant chances of accumulating in the food chain.

Phenolic compounds

Phenols and their derivatives, i.e. chlorophenol are distributed commonly in the environment.

Such compounds are mainly used in the production of dyes, drugs, plastics, pesticides, detergents, etc..

IN food industry:

- Food processing industry faces various challenges; one of the foremost challenges is the need for quick and cost effective methods to detect the presence of allergenic components and pathogens in the food.
- Biosensors pave way for the rapid detection of pathogens, allergens as well as pesticide residues in food.
- Detection of contaminants, verification of product contents, product freshness and monitoring of raw materials conversion are the areas of potential biosensor applications.

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- Biosensors have the potential to produce an analytical revolution to resolve the challenges in the agricultural and the food industries.
- The determination of chemical and biological contaminants in foods is of paramount importance to the health of food because, unlike the contamination of a physical nature, they cannot be displayed.
- Development of biosensors to the analysis of the quality of food, since they have proven to be an extremely viable alternative to traditional analytical techniques such as chromatography.
- In the area of food the interest in the development of biosensors mainly focuses on analysis of food security (detection of compound contaminants, allergens, toxins, pathogens, and additives etc.) Food composition and online process control.

Biosensors for diabetes applications

Glucose as diabetes biomarker

- About 3% of the population worldwide suffers from diabetes, a leading cause of death, and its incidence is growing fast.
- Diabetes is a syndrome of disordered metabolism resulting in abnormally high blood sugar levels.
- Optimal management of diabetes involves patients measuring and recording their own blood glucose levels.
- Under normal physiological condition, the concentration of fasting plasma glucose is in the range 6.1–6.9 mmolL⁻¹, so the variation of the blood glucose level can indicate diabetes mellitus, besides other conditions.
- Association recommends that insulin-dependent type 1 diabetics self monitor blood glucose 3–4 times daily, while insulin-dependent type 2 diabetics monitor once-daily.
- Therefore it is necessary to develop a simple, sensitive, accurate, micro-volume and
- low-cost approach for glucose analysis which is appropriate for rapid field tests and is also effective as an alternative to the existing methods.

First-generation of glucose biosensors (Clark enzyme electrode)

- First proposed in 1962 by Clark and Lyons
- These sensors were initially based on an electrochemical approach, which used the enzyme glucose oxidase (GOx).
- Electrochemical sensors were chosen for blood-glucose measurements due to their high sensitivity, on the order of μM to mM, good reproducibility and ease of fabrication at relatively low cost.
- GOx catalyses the oxidation of glucose to gluconolactone in the presence of oxygen, while producing hydrogen peroxide (H_2O_2) and water as by-products.

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- Measurements of peroxide formation have the advantage of being simpler, especially when miniature devices are being considered.
- The first commercially successful glucose biosensor using Clark's technology was the Yellow Springs Instrument Company analyzer (Model 23A YSI analyzer) for the direct measurement of glucose in 1975, which was based on the amperometric detection of hydrogen peroxide.
- This analyzer was almost exclusively used in clinical laboratories because of its high cost due to the expensive platinum electrode.

BT405

Quantum Continent

Role of Nanotechnology in Diagnostics:

Several nanoparticles have been used for diagnostics. Of these, the most frequently used are gold nanoparticles, QDs, and magnetic nanoparticles.

Nanodiagnostics is the new term that describes use of methods and techniques of nanotechnology and its principles for diagnostics purposes.

It includes, although not limited to, the manipulation and assessment of single molecule, size reduction of systems and platforms to make use of nanoscale properties obtainable from interactions between surfaces and biomolecules.

Nanodiagnostics is an evolving application of nanoscale technology to meet the demand of clinical diagnostics, determining disease state, any predisposition to such, the pathology of the condition and the identification of causative organisms

Five uses of nanotechnology in daily life

Food & packaging

Textile and clothing

Sports & leisure, household

Computing & electronics

Transport, building, construction

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Medical

Environment The total market for nanobiotechnology products was \$19.3 billion in 2010 and is growing at a compound annual growth rate (CAGR) of 9% to reach a forecast market size of \$29.7 billion by 2015.

Medical applications, including drug delivery and microbicides, dominate today's market, with sales of \$19.1 billion in 2010. This market segment is growing at a compound annual growth rate (CAGR) of 8.7%, and is forecast to reach sales of \$29 billion by 2015 (BCC).

More and more consumer products are branded worldwide as nano.

The most common nanoscale substance used in consumer products is nano-silver (anti-bacterial and anti-fungal properties)

Baby bottle nipples, door handles, food contact products, interiors of refrigerators and washing machines, clothing, bedding, paints, cleaning products, food storage containers, bandages, medical devices, mattresses, and children's teddy bears

Spectroscopy and X-ray siRNA Drug Delivery System

Among these clinical trials, most siRNAs were administered by local delivery, typically via the [intravitreal](#) or [intranasal routes](#). However, local delivery may not be appropriate for all diseases. Under some circumstances, systemic drug administration by [intravenous](#) Compared to chemotherapeutic anti-cancer drugs, there are a lot of advantages of siRNA drug. Due to the special mechanism of siRNA, it has four advantages as a potential cancer therapeutic strategy. The first is its high degree of safety. siRNA acts on the post-translational stage of gene expression, so it does not interact with DNA and thereby avoids the mutation and teratogenicity risks of gene therapy. The second advantage of siRNA is its high efficacy. In a single cancer cell, siRNA can cause dramatic suppression of gene expression with just several copies.

Types of Glucose Biosensors

Types of Glucose Biosensors

Enzymatic glucose biosensors

First generation glucose biosensor

Second generation glucose biosensor

Third generation glucose biosensor

Non-enzymatic glucose biosensors

First-generation of glucose biosensors (Clark enzyme electrode)

First proposed in 1962 by Clark and Lyons

These sensors were initially based on an electrochemical approach, which used the enzyme glucose oxidase (GOx).

Electrochemical sensors were chosen for blood-glucose measurements due to their high sensitivity, on the order of μM to mM , good reproducibility and ease of fabrication at relatively low cost.

GOx catalyses the oxidation of glucose to gluconolactone in the presence of oxygen, while producing hydrogen peroxide (H_2O_2) and water as by-products.

Second-generation of Glucose Biosensors (Mediated Biosensors)

Limitations of the first generation glucose biosensors were overcome by using mediated glucose biosensors, second generation glucose sensors.

The improvements were achieved by replacing oxygen with non-physiological electron acceptors, called redox mediators that were able to carry electrons from the enzyme to the surface of the working electrode.

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Third-generation of Glucose Biosensors

This feature allows a 3rd generation glucose sensing mechanism, which is independent of mediators or oxygen.

The third-generation glucose biosensors are reagentless and based on direct transfer between the enzyme and the electrode without mediators.

Instead of mediators with high toxicity, the electrode can perform direct electron transfers using organic conducting materials based on charge-transfer complexes.

Optical Microscope and its resolution power

- An optical microscope uses visible light (i.e. electromagnetic radiation) and a system of lenses to magnify images of small samples.
- For this reason, it is also called a light microscope. Optical microscopes are the oldest and simplest of the microscopes.
- The resolution limit of an optical microscope is governed by the wavelength of visible light.
- Visible light is the part of the electromagnetic spectrum with wavelengths between 400 and 700 nm.
- The resolving power of an optical microscope is around 0.2 μm or 200 nm: thus, for two objects to be distinguishable, they need to be separated by at least 200 nm.

Single objects smaller than this limit are not distinguishable: they are seen as fuzzy objects. This is known as the diffraction limit of visible light.

Microarray

Application of Biosensors

- Other microscopes have been designed which use other beams: rather than light, they use electron beams to illuminate the sample.
- Electron microscopes have much greater resolving power than light microscopes that use electromagnetic radiation and can obtain much higher magnifications of up to two million times, while the best light microscopes are limited to magnifications of 2 000 times.
- There are various types of electron microscopes, such as the scanning electron microscope (SEM) and or the transmission electron microscope (TEM).

Scanning Tunnelling Microscope (STM) can create detailed 3D images of a sample with atomic resolution.